MATH 8090:
State-Space Models and Geostatistics

Readings: SS17 Chapters 6.2–6.4 and 6.12; BD Chapters 9.4–9.7

Whitney Huang



School of
MATHEMATICAL AND
STATISTICAL SCIENCES
Clemson University

Week 14

Review: Kalman Recursions for Filtering/Forecasting

1. Compute innovation:

$$U_t = Y_t - \hat{Y}_{t|t-1} = Y_t - \mu_t^f$$

2. Compute MSE for $\hat{Y}_{t|t-1}$:

$$\Sigma_t^f + \sigma_W^2 = F_t$$

3. Compute new filtered value:

$$\mu_t^a = \mu_t^f + K_t U_t,$$

where $K_t = \Sigma_t^f / F_t$ is the so-called Kalman gain

4. Compute MSE for new filtered value:

$$\Sigma_t^a = \Sigma_t^f (1 - K_t)$$

Kalman Recursions for Filtering/Forecasting (Cont'd)

5. Compute new forecast:

$$\mu_{t+1}^f = \mu_t^f + K_t U_t = \mu_t^a$$

6. Compute MSE for new forecast:

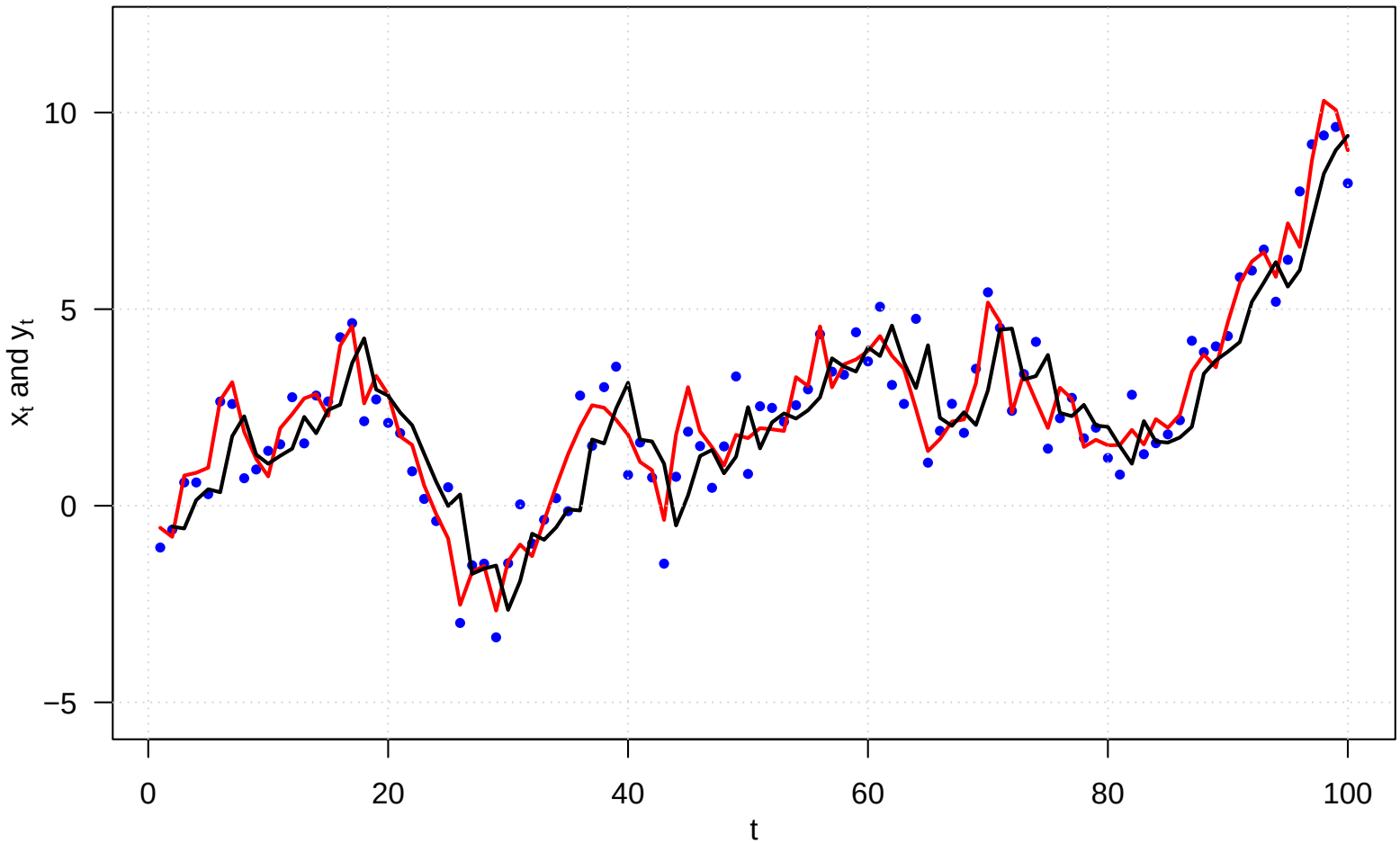
$$\Sigma_{t+1}^f = \Sigma_t^f (1 - K_t) + \sigma_V^2 = \Sigma_t^a + \sigma_V^2$$

Recursions are carried out for $t = 0, \cdots, n$ with inputs $\mathbb{E}[X_0] = \mu_0$, $\mathrm{Var}(X_0) = \sigma_0^2$ and $Y_t's$

Simulated Example: Local Level Model with SNR $= 2$

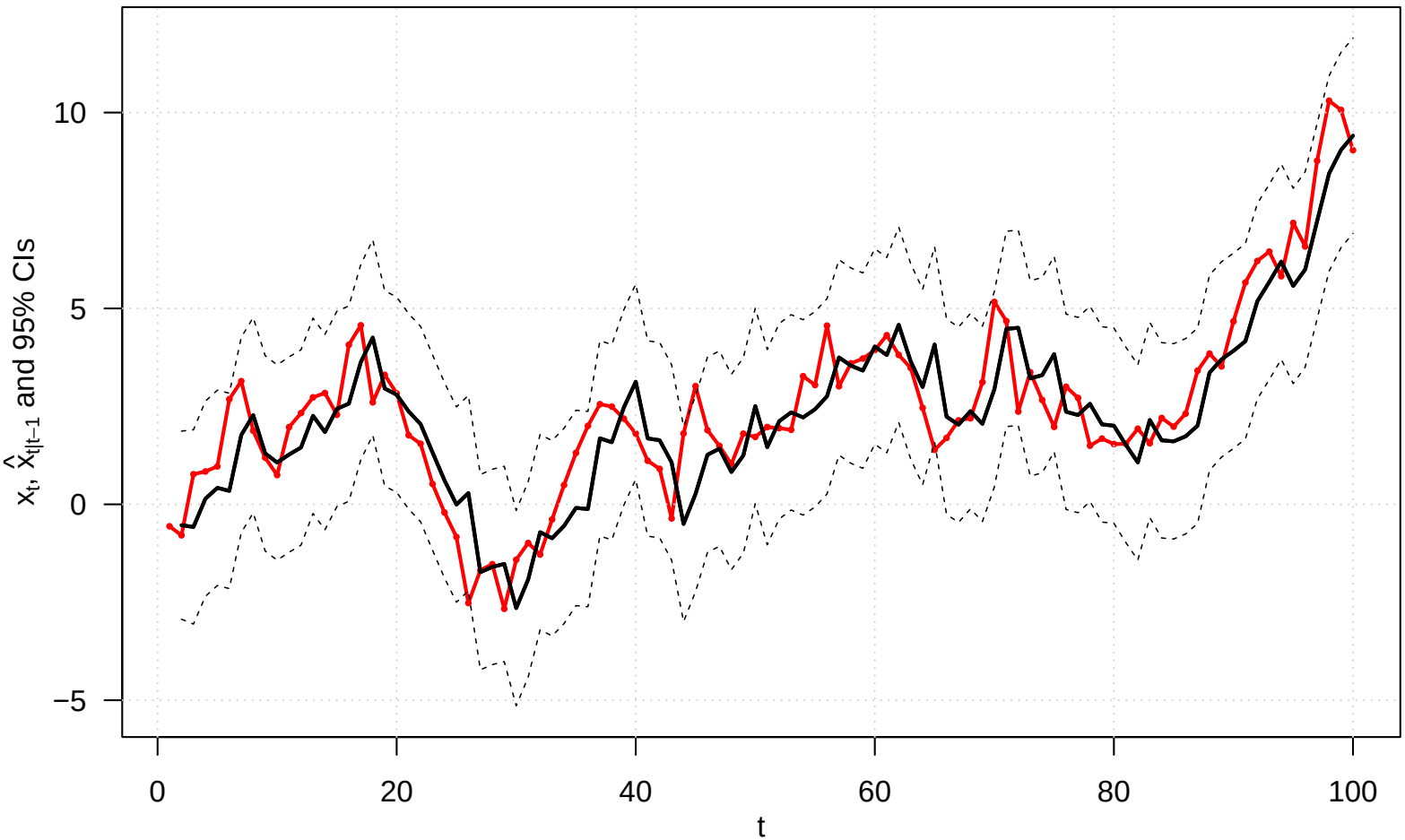
Setup: $\mu_0 = 0$, $\sigma_0^2 = 1 = \sigma_V^2$, $\sigma_W^2 = 0.5$

Time series Y_t, states X_t, and forecasts μ_t^f



Simulated Data from Local Level Model with SNR $= 2$

States X_t, forecasts μ_t^f, and 95% CIs based on Σ_t^f



Kalman Filtering with Missing Observations

Suppose Y_{t+1} and Y_{t+2} are missing, but Y_{t+3} is observed

- When Y_j is missing, skip the measurement update:

$$\mu_j^a = \mu_j^f, \quad \Sigma_j^a = \Sigma_j^f.$$

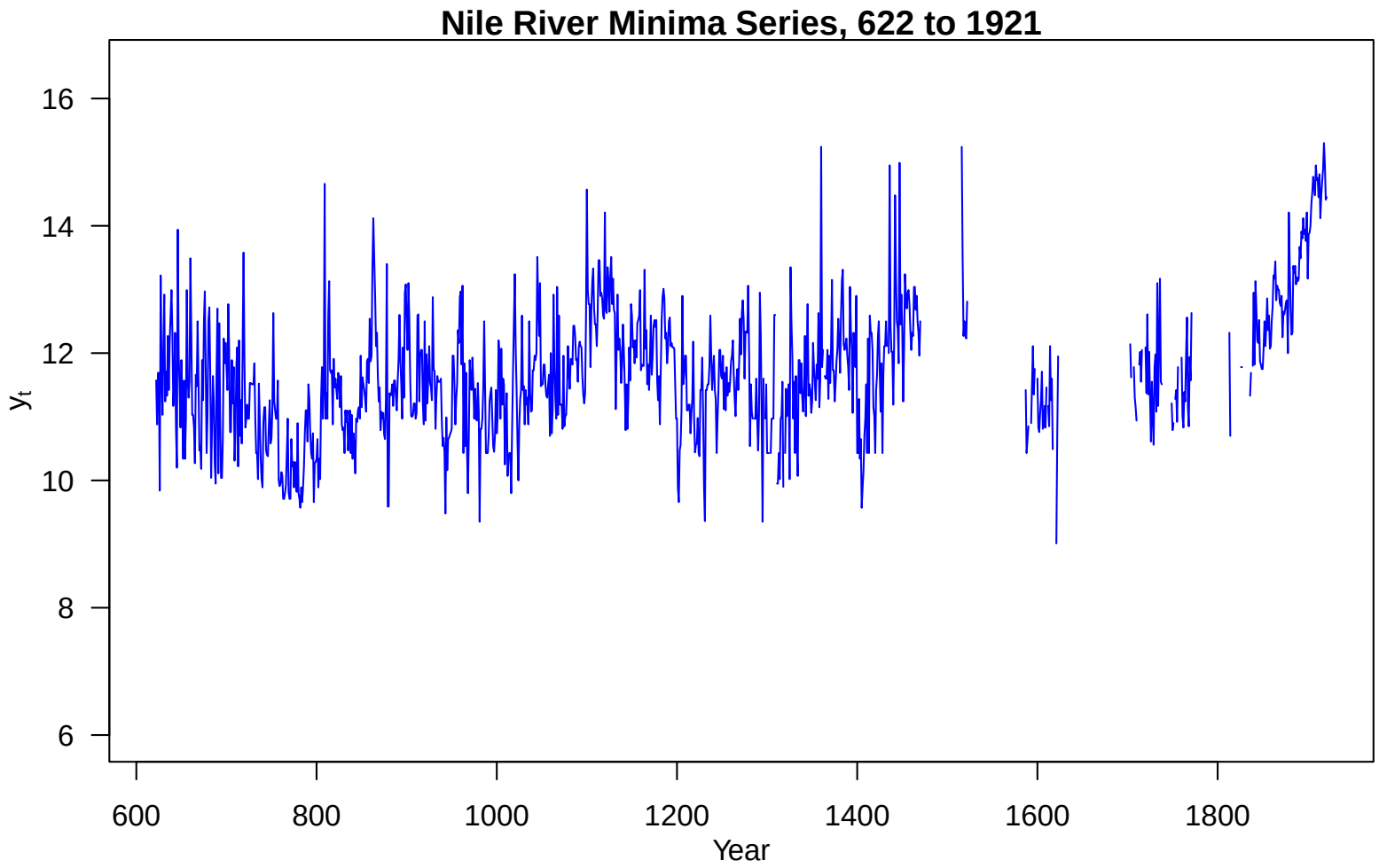
- Continue the state forecast (local-lelve model):

$$\mu_{j+1}^f = \mu_j^a, \quad \Sigma_{j+1}^f = \Sigma_j^a + \sigma_V^2.$$

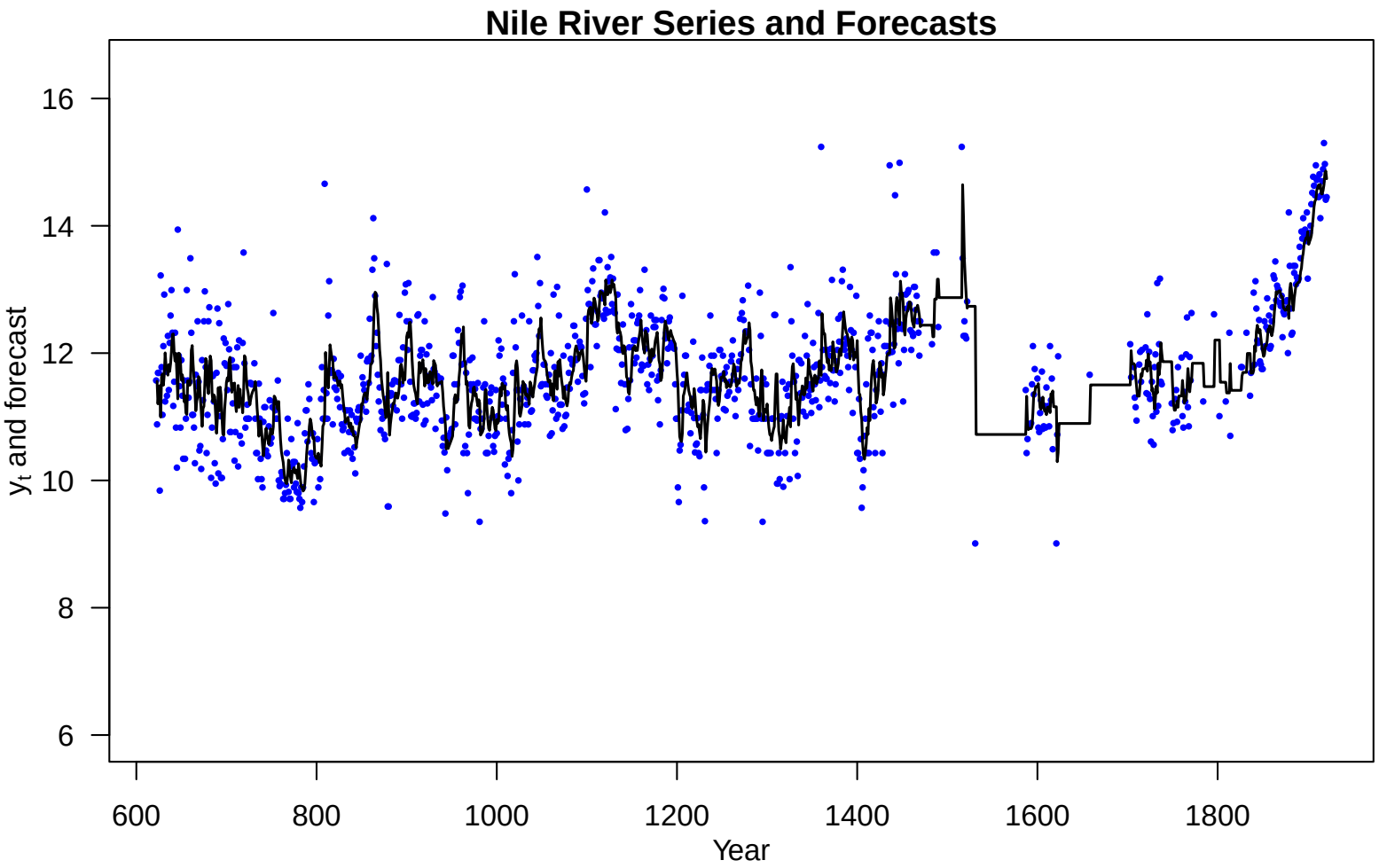
- When Y_{t+3} arrives, resume the usual innovation and update steps.

Key points: Missing data do not interrupt state propagation; uncertainty grows without measurement updates

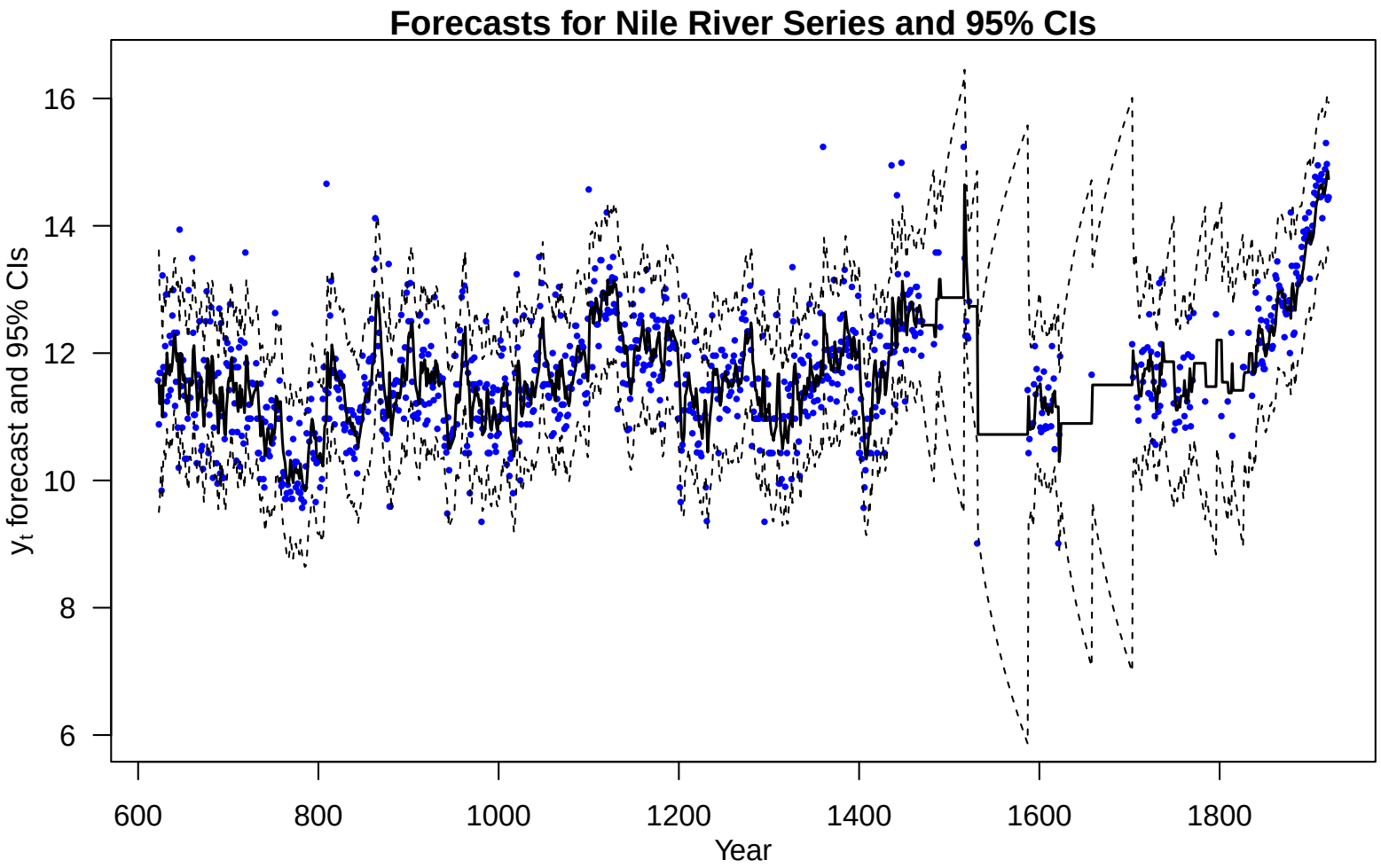
Example: Nile River Annual Minima Series



Nile River Series with Missing Values Imputed



Nile River Series Forecasts with 95 % CI



Kalman Recursions for Smoothing: I

Given time series $Y_1, \cdots, Y_n$, Kalman filter recursions give us $\mu_t^a = \mathbb{E}[X_t | \mathbf{Y}_{1:t}]$ for $t = 1, \cdots, n$

- Regression lemma says solution to smoothing problem is

$$\mu_t^s = \mathbb{E}[X_t | \boldsymbol{Y}_{1:n}] = \mu_t + \Sigma_{t,n}^T \Sigma_{Y,n}^{-1} (\boldsymbol{Y}_{1:n} - \boldsymbol{\mu}_{1:n})$$

- MSE for predictor, i.e., $\mathbb{E}\left[(X_t - \mu_t^s)^2\right]$, is

$$\Sigma_t - \Sigma_{t,n}^T \Sigma_{Y,n}^{-1} \Sigma_{t,n} = \Sigma_t^s,$$

where $\Sigma_t = \mathrm{Var}[X_t]$

Kalman Recursions for Smoothing: II

Using innovation U_t, innovation variance F_t, Kalman gains K_t, forecasts μ_t^f and associated MSEs $\Sigma_t^f, t = 1, \cdots, n$ computed by Kalman filter recursions, Kalman smoother recursions allow efficient computation of $\mu_t^s, t = 1, \cdots, n$

The first two steps yield desired predictor μ_t^s

1. Manipulate innovations: starting with $r_n = 0$, recursively form
$$r_{t-1} = \frac{U_t}{F_t} + (1 - K_t) r_t, \quad t = n, \cdots, 1$$

2. Combine manipulated innovations and forecasts:
$$\mu_t^s = X_t^t + \Sigma_t^f r_{t-1}, \quad t = 1, \cdots, n$$

Kalman Recursions for Smoothing: III

Next two steps yield MSE for predictor μ_t^s:

3. Manipulate innovation variances: starting with $N_n = 0$, recursively form

$$N_{t-1} = \frac{1}{F_t} + (1 - K_t)^2 N_t, \quad t = n, \cdots, 1$$

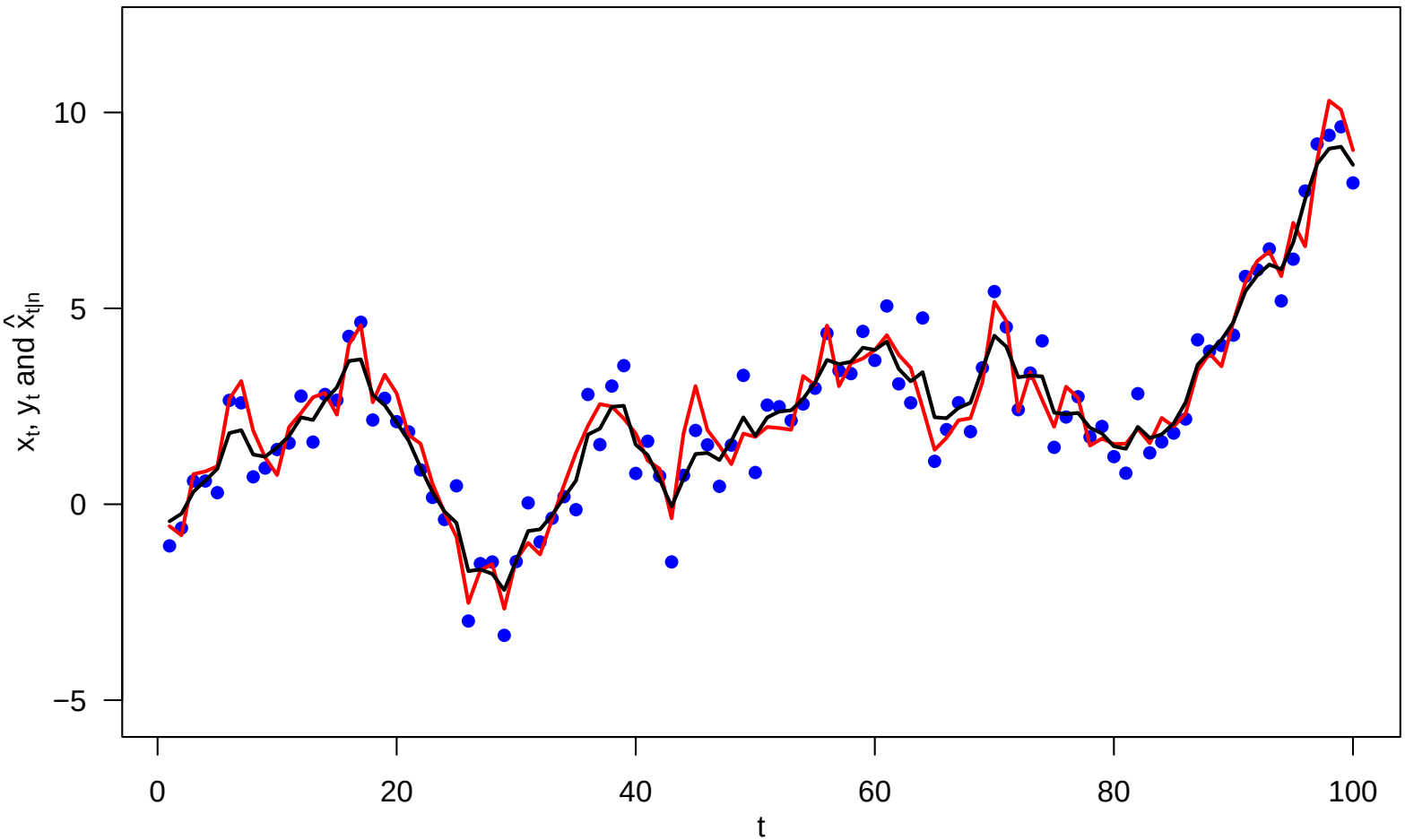
4. Combine manipulated innovation variances and forecast MSEs:

$$\Sigma_t^s = \Sigma_t^f - \left(\Sigma_t^f\right)^2 N_{t-1}, \quad t = 1, \cdots, n,$$

where Σ_t^s is the desired MSE

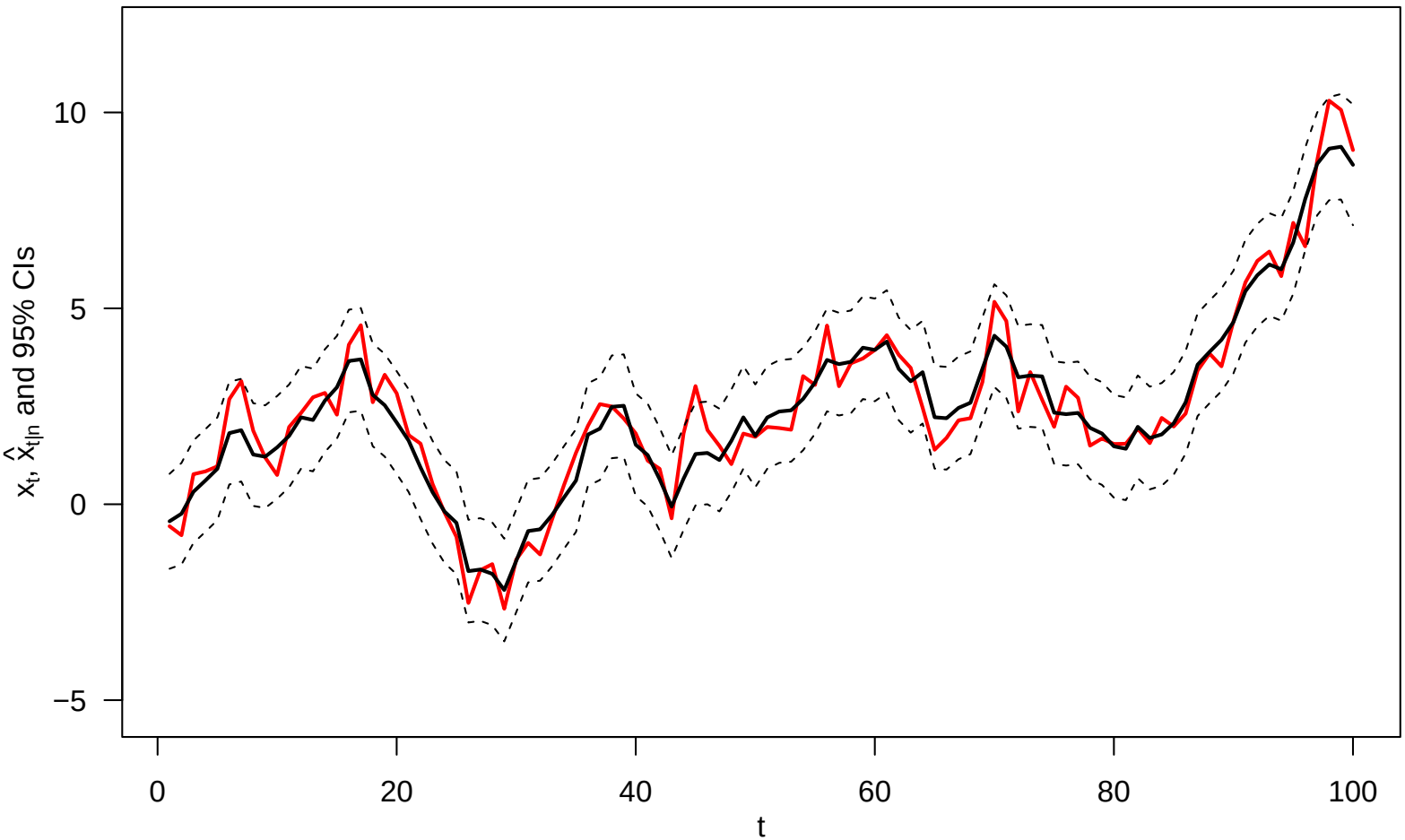
Simulated Example: Local Level Model with SNR $= 2$

Time series Y_t, states X_t, and smooths μ_t^s



Simulated Data from Local Level Model with SNR $= 2$

States X_t, smooths X_t^n, and 95% CIs based on Σ_t^s



Estimating the State-Space Model Parameters

So far, we've assumed that the parameters $\boldsymbol{\theta} = (\sigma_V^2, \sigma_W^2, \mu_0, \sigma_0^2)$ are known. In practice, we need to estimate from the data

This requires maximizing the marginal likelihood of the data $\boldsymbol{y}$, having integrated the latent time series $\boldsymbol{x}$ out. This is given by:

$$f(\mathbf{y}|\sigma_V^2, \sigma_W^2, \mu_0, \sigma_0^2) = \int f(\mathbf{y}|\mathbf{x}, \sigma_W^2) f(\mathbf{x}|\mu_0, \sigma_0^2, \sigma_V^2)\, d\mathbf{x}$$

Maximizing over an integral can be difficult ☹

Direct Maximum Marginal Likelihood

Under the Gaussian state-space model, the marginal distribution of $\mathbf{y}$ is

$$\mathbf{y} \sim \mathrm{N}(\mathbb{E}(\mathbf{x}), \mathrm{Var}(\mathbf{x}) + \sigma_W^2 I_n).$$

Direct evaluation involves an $n \times n$ covariance matrix.

Innovations likelihood:

$$U_t = Y_t - \mathbb{E}[Y_t | \mathbf{Y}_{1:t-1}], \quad F_t = \mathrm{Var}(U_t | \mathbf{Y}_{1:t-1}).$$

The Kalman filter evaluates the log-likelihood recursively:

$$\boxed{\ell(\boldsymbol{\theta}) = -\frac{1}{2} \sum_{t=1}^{n} \left\{ \log(2\pi) + \log F_t + \frac{U_t^2}{F_t} \right\}.}$$

- For each candidate $\boldsymbol{\theta}$, run the filter to obtain U_t and F_t.
- Numerically maximize $\ell(\boldsymbol{\theta})$ over valid parater values.

Maximum Marginal Likelihood via EM

Another way to compute maximum likelihood estimate $\hat{\boldsymbol{\theta}}$ is to use the expectation-maximization (EM) algorithm [Dempster, Laird, and Rubin, 1977]

- Initialize by choosing starting value $\boldsymbol{\theta}^0$, and compute the incomplete likelihood
- Perform the E-step to obtain μ_t^s, Σ_t^s
- Perform M-step to update the estimate $\boldsymbol{\theta}$ using the complete likelihood
- Recompute the incomplete likelihood
- Repeat until convergence, i.e., $|\hat{\boldsymbol{\theta}}^N - \hat{\boldsymbol{\theta}}^{N-1}| < \epsilon$

Bayesian Estimation

Markov Chain Monte Carlo (MCMC) methods, such as the Gibbs sampler [Gelfand and Smith, 1990] or the Metropolis-Hastings algorithm [Metropolis et al., 1953; Hastings, 1970], are commonly used for Bayesian inference in state space models

Gibbs Sampler for State Space Models

1. Draw $\boldsymbol{\theta}$ from $p(\boldsymbol{\theta}|\mathbf{x}_{0:n}, \mathbf{y_{1:n}})$, where

$$p(\boldsymbol{\theta}|\mathbf{x}_{0:n}, \mathbf{y}_{1:n}) \propto \pi(\boldsymbol{\theta}) p(x_0|\boldsymbol{\theta}) \prod_{t=1}^{n} p(x_t|x_{t-1}, \boldsymbol{\theta}) p(y_t|x_t, \boldsymbol{\theta})$$

2. Draw $\mathbf{x}_{0:n}$ from $p(\mathbf{x}_{0:n}|\mathbf{y}_{1:n}, \boldsymbol{\theta})$, where

$$p(\mathbf{x}_{0:n}|\mathbf{y}_{1:n}, \boldsymbol{\theta}) = p(x_n|\boldsymbol{y}_{1:n}, \boldsymbol{\theta}) p(x_{n-1}|x_n, \mathbf{y}_{1:n-1}, \boldsymbol{\theta}) \cdots p(x_0|x_1, \boldsymbol{\theta})$$

Use forward-filtering, backward sampling (FFBS) algorithm to sequentially simulate the individual states backward

From Kalman Filtering to Spatial Prediction

A common principle: Gaussian conditioning

- Time series: infer an unobserved state X_t from noisy measurements $\mathbf{Y}_{1:t}$.
- Spatial data: infer $Y(\mathbf{s}_0)$ at an unobserved location from measurements at $\mathbf{s}_1, \cdots, \mathbf{s}_n$.

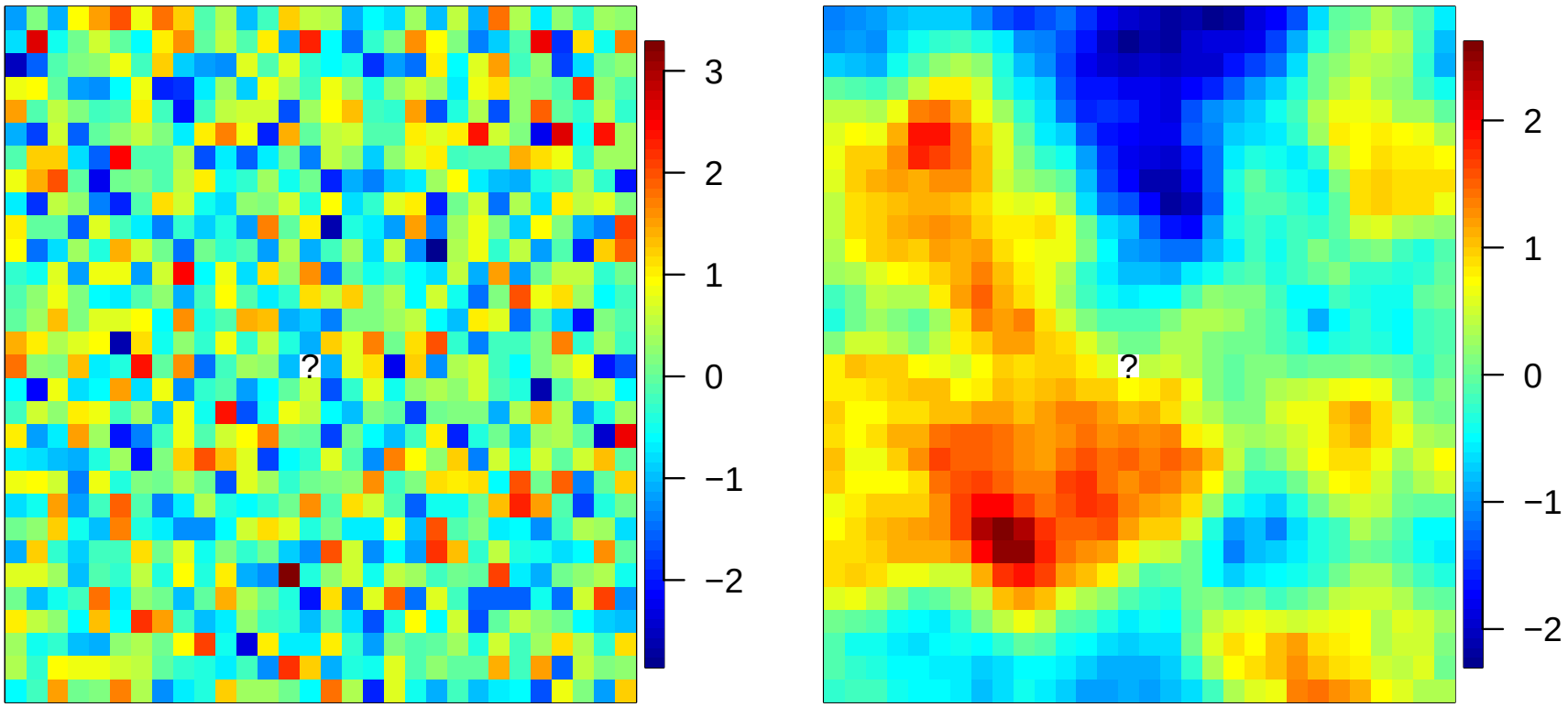
Both rely on **mean and covariance** to combine information and quantify uncertainty

$$\boxed{\text{Kalman filter (time)} \rightleftharpoons \text{Kiging (space)}}$$

Geostatistics

Toy Examples of Spatial Interpolation

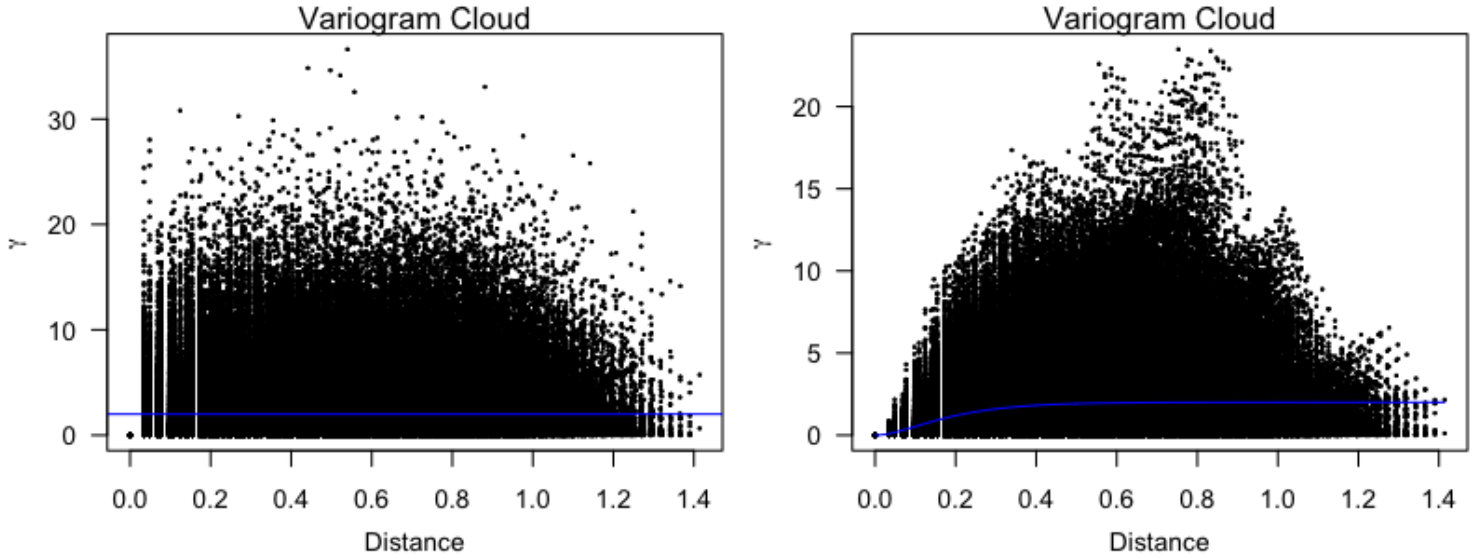
Let's consider two spatial images, each with a missing pixel

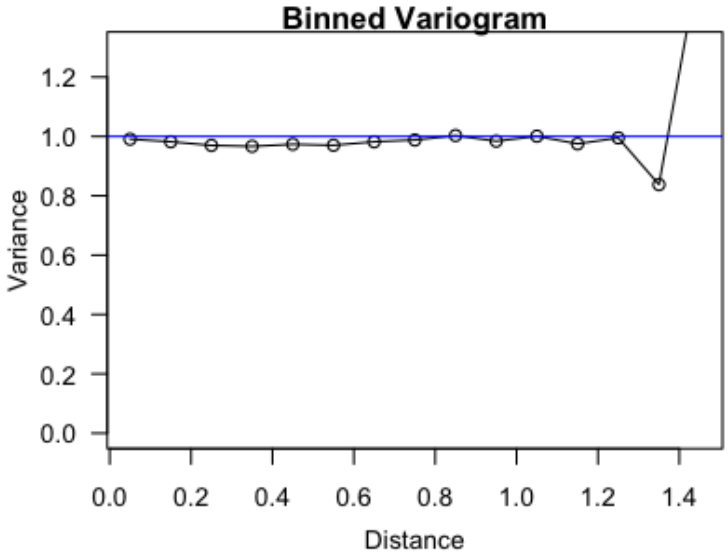


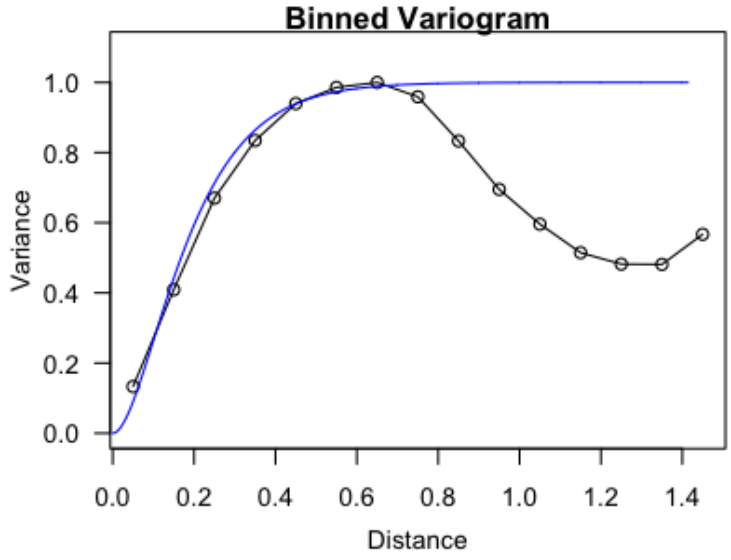
Question: What is your best guess of the value of the missing pixel, denoted as $Y(\mathbf{s}_0)$, for each case?

Visualizing Spatial Dependence Structure

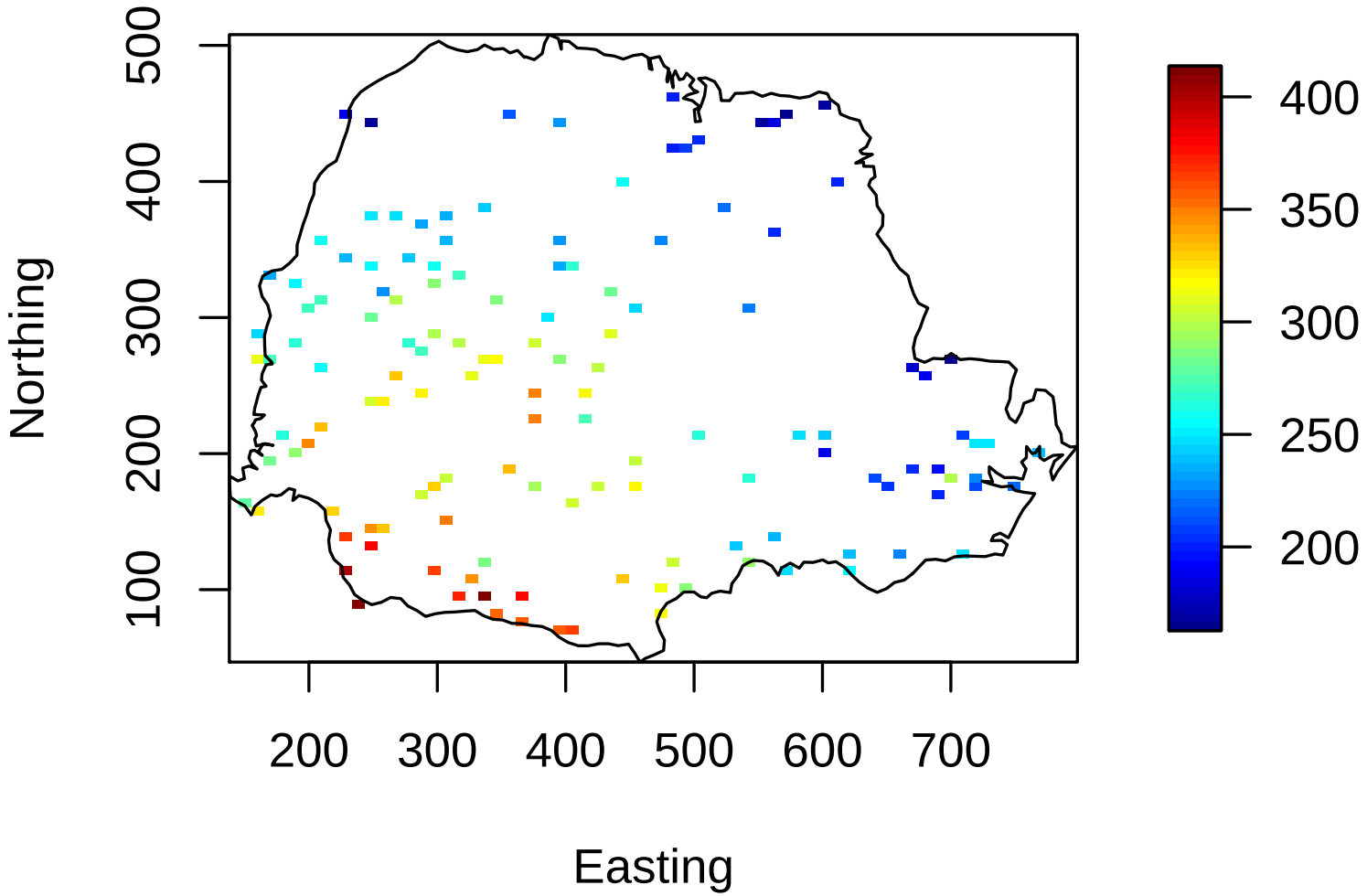
Closer locations are more alike; the **variogram** measures how similarity decays with distance $\gamma(h) = \frac{1}{2}\mathrm{Var}(Y(s) - Y(s+h))$







Interpolating Paraná State Precipitation Data



Goal: To interpolate the values in the spatial domain

The Spatial Interpolation Problem

Given observations of a spatially varying quantity Y at n spatial locations

$$y(\mathbf{s}_1), y(\mathbf{s}_2), \cdots, y(\mathbf{s}_n), \qquad \mathbf{s}_i \in \mathcal{S},\ i = 1, \cdots, n$$

We want to estimate this quantity at any **unobserved location**

$$Y(\mathbf{s}_0), \qquad \mathbf{s}_0 \in \mathcal{S}, \quad \hat{Y}(\mathbf{s}_0) = \mathbb{E}[Y(\mathbf{s}_0) \mid \mathbf{Y} = \mathbf{y}]$$

$\Rightarrow$ The optimal predictor under **squared error loss** is the **conditional expectation**

Applications

- Mining: ore grade
- Climate: temperature, precipitation, $\cdots$
- Environmental Science: air pollution levels, $\cdots$

A Brief History of Spatial Statistics

Applied Origins

Mining, geology, and forestry: interpolation and resource estimation [Krige, 1951].

Geostatistics

Variograms [Matheron, 1960s] and covariance models [Matérn, 1960].

Statistical Foundations

Systematic development of spatial modeling and prediction [Cressie, 1993; Stein, 1999].



Krige



Matheron



Matérn



Cressie



Stein

Spatial statistics grew from geoscience applications into a statistical discipline grounded in stochastic processes and spatial prediction.

Outline

Kalman Recursions for Filtering, Forecasting, and Smoothing

Estimating the State-Space Model Parameters

Gaussian Process Spatial Model

Spatial Interpolation

Parameter estimation

A Case Study of Paraná State Precipitation Data

Linear Interpolation

The best guess (under squared loss) should be based on the conditional distribution $[Y(\mathbf{s}_0) | \mathbf{Y} = \mathbf{y}]$ where

$$\mathbf{y} = (y(\mathbf{s}_1), \cdots, y(\mathbf{s}_n))^{\mathrm{T}}$$

- Calculating this conditional distribution can be difficult
- Instead we use a linear predictor:

$$\hat{Y}(\mathbf{s}_0) = \lambda_0 + \sum_{i=1}^{n} \lambda_i y(\mathbf{s}_i)$$

- The best linear predictor is completely determined by the mean and covariance of $\{Y(\boldsymbol{s}),\, \boldsymbol{s} \in \mathcal{S}\}$

Why Linear?

Linear predictors are

- Computationally simple
- Optimal for Gaussian processes

Gaussian Process (GP) Spatial Model

Suppose the observed data $\{Y(\mathbf{s}_i)\}_{i=1}^n$ are measurements from an underlying spatial process $\{Y(\mathbf{s})\}_{\mathbf{s}\in\mathcal{S}}$, where $\mathcal{S} \subset \mathbb{R}^d$ denotes the spatial domain.

$$\text{Model:} \qquad Y(\mathbf{s}) = m(\mathbf{s}) + \epsilon(\mathbf{s}), \qquad \mathbf{s} \in \mathcal{S}.$$

- Mean function (trend): $m(\mathbf{s}) = \mathrm{E}[Y(\mathbf{s})] = \mathbf{x}^T(\mathbf{s})\boldsymbol{\beta}$, which captures the large-scale spatial variation through spatial covariates
- Spatial residual process: $\{\epsilon(\mathbf{s})\}_{\mathbf{s}\in\mathcal{S}} \sim \mathrm{GP}(0, K(\cdot,\cdot))$, where
$$K(\mathbf{s}_1, \mathbf{s}_2) = \mathrm{Cov}(\epsilon(\mathbf{s}_1), \epsilon(\mathbf{s}_2))$$
is the covariance function describing spatial dependence
- Nearby locations tend to have similar residual values, with dependence decreasing as distance increases.

Assumptions on Covariance Function

In practice, the covariance must be estimated from the data $(y(\mathbf{s}_1), \cdots, y(\mathbf{s}_n))^{\mathrm{T}}$. We need to impose some structural assumptions

- Stationarity:

$$\begin{aligned} K(\mathbf{s}_1, \mathbf{s}_2) &= \mathrm{Cov}\left(\epsilon(\mathbf{s}_1), \epsilon(\mathbf{s}_2)\right) = C(\mathbf{s}_1 - \mathbf{s}_2) \\ &= \mathrm{Cov}\left(\epsilon(\mathbf{s}_1 + \mathbf{h}), \epsilon(\mathbf{s}_2 + \mathbf{h})\right)) \end{aligned}$$

- Isotropy:

$$K(\mathbf{s}_1, \mathbf{s}_2) = \mathrm{Cov}\left(\epsilon(\mathbf{s}_1), \epsilon(\mathbf{s}_2)\right) = C(\|\mathbf{s}_1 - \mathbf{s}_2\|)$$

Key message: Stationarity makes covariance depend on separation rather than absolute location; isotropy further removes directional dependence, leaving only distance.

A Valid Covariance Function Must Be Positive Definite!

A covariance function is positive definite (p.d.) if

$$\sum_{i,j=1}^{n} a_i a_j C(\boldsymbol{s}_i - \boldsymbol{s}_j) \geq 0$$

for any finite locations $\boldsymbol{s}_1, \cdots, \boldsymbol{s}_n$, and for any constants a_i, $i = 1, \cdots, n$

Question: what is the consequence if a covariance function is NOT p.d.? $\Rightarrow$ We can get a negative variance

Question: How to guarantee a $C(\cdot)$ is p.d.?

- Using a parametric covariance function (see some examples in next slide)
- Using Bochner's Theorem to construct a valid covariance function

Bochner's Theorem

When is a stationary covariance function valid?

A continuous function $C : \mathbb{R}^d \to \mathbb{C}$ is a stationary covariance function if and only if

$$\boxed{C(\boldsymbol{h}) = \int_{\mathbb{R}^d} e^{i\boldsymbol{\omega}^T \boldsymbol{h}} \, F(d\boldsymbol{\omega}),}$$

where F is a finite nonnegative **spectral measure**.

If $F(d\boldsymbol{\omega}) = f(\boldsymbol{\omega}) \, d\boldsymbol{\omega}$, then f is the **spectral density**. If C is integrable,

$$f(\boldsymbol{\omega}) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\boldsymbol{\omega}^T \boldsymbol{h}} C(\boldsymbol{h}) \, d\boldsymbol{h}.$$

Key implication: Any nonnegative, integrable spectral density defines a valid stationary covariance. For real-valued fields, also require $f(\boldsymbol{\omega}) = f(-\boldsymbol{\omega})$.

Some Commonly Used Covariance Functions

- Powered exponential:

$$C(h) = \sigma^2 \exp\left(-(\frac{h}{\rho})^\alpha\right), \qquad \sigma^2 > 0,\ \rho > 0,\ 0 < \alpha \leq 2$$

- Spherical:

$$C(h) = \sigma^2 \left(1 - 1.5\frac{h}{\rho} + 0.5\left(\frac{h}{\rho}\right)^3\right) 1_{\{h \leq \rho\}}, \qquad \sigma^2,\ \rho > 0$$

  Note: it is only valid for 1,2, and 3 dimensional spatial domain.

- Matérn:

$$C(h) = \sigma^2 \frac{\left(\sqrt{2\nu}h/\rho\right)^\nu \mathcal{K}_\nu\left(\sqrt{2\nu}h/\rho\right)}{\Gamma(\nu)2^{\nu-1}}, \qquad \sigma^2 > 0,\ \rho > 0,\ \nu > 0$$

"Use the Matérn model" – Stein (1999, pp. 14)

1-D Realizations from Matérn Model with Fixed σ^2, ρ

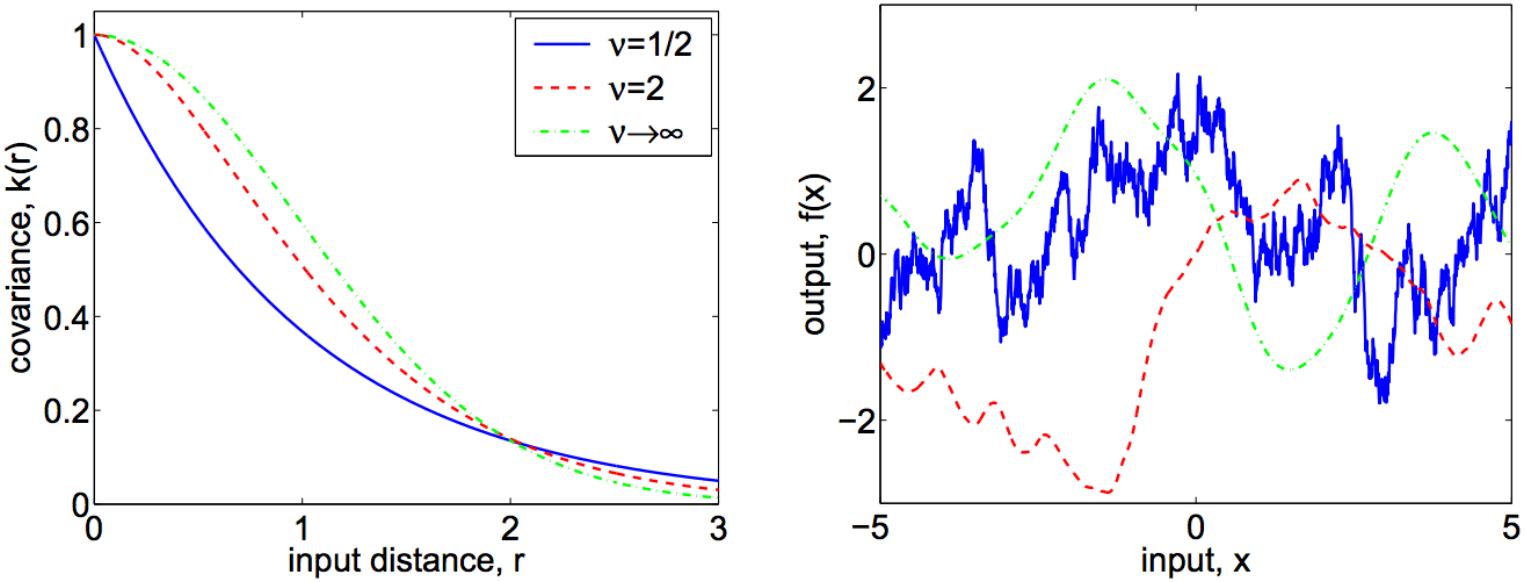
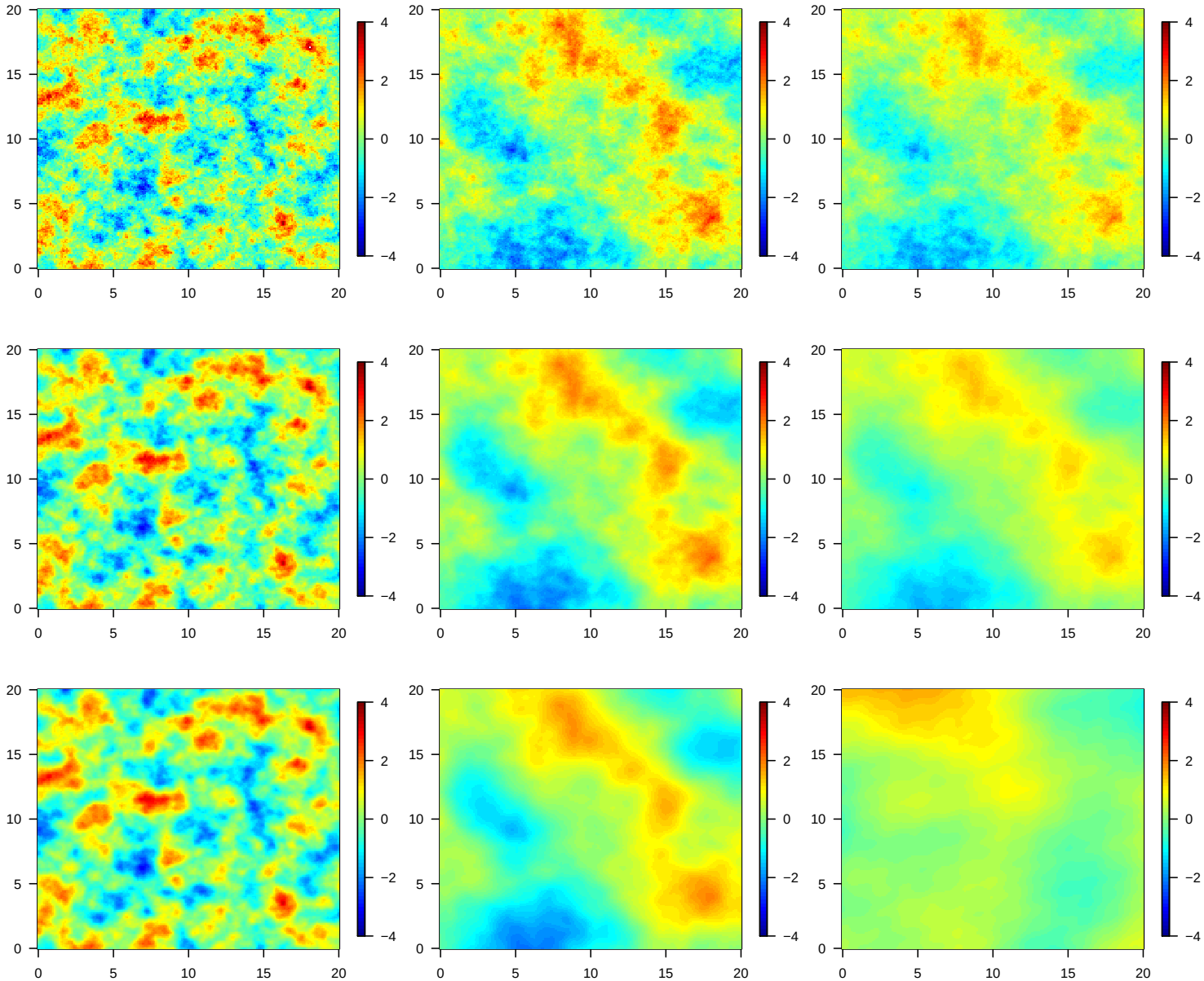


Figure: courtesy of Rasmussen & Williams 2006

The larger ν is, the smoother the process is

2-D Realizations from Matérn Model with Fixed σ^2



Outline

Kalman Recursions for Filtering, Forecasting, and Smoothing

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Gaussian Process Spatial Model

Spatial Interpolation

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A Case Study of Paraná State Precipitation Data

Conditional Distribution of Multivariate Normal

If

$$\begin{pmatrix} \mathbf{Y}_1 \\ \mathbf{Y}_2 \end{pmatrix} \sim \mathrm{N}\left(\begin{pmatrix} \boldsymbol{\mu_1} \\ \boldsymbol{\mu_2} \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right)$$

Then

$$[\mathbf{Y}_1 | \mathbf{Y}_2 = \mathbf{y}_2] \sim \mathrm{N}\left(\boldsymbol{\mu_{1|2}}, \Sigma_{1|2} \right)$$

where

$$\begin{aligned} \boldsymbol{\mu_{1|2}} &= \boldsymbol{\mu_1} + \Sigma_{12}\Sigma_{22}^{-1}\left(\mathbf{y}_2 - \boldsymbol{\mu_2}\right) \\ \Sigma_{1|2} &= \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} \end{aligned}$$

GP-Based Spatial Interpolation: Kriging

If $\{Y(\mathbf{s})\}_{\mathbf{s}\in\mathcal{S}}$ follows a GP, then

$$\begin{pmatrix} Y_0 \\ \boldsymbol{Y} \end{pmatrix} \sim \mathrm{N}\left(\begin{pmatrix} m_0 \\ \boldsymbol{m} \end{pmatrix}, \begin{pmatrix} \sigma_0^2 & k^{\mathrm{T}} \\ k & \Sigma \end{pmatrix} \right)$$

We have

$$[Y_0|\boldsymbol{Y}=\boldsymbol{y}] \sim \mathrm{N}\left(m_{Y_0|\boldsymbol{Y}=\boldsymbol{y}}, \sigma^2_{Y_0|\boldsymbol{Y}=\boldsymbol{y}}\right)$$

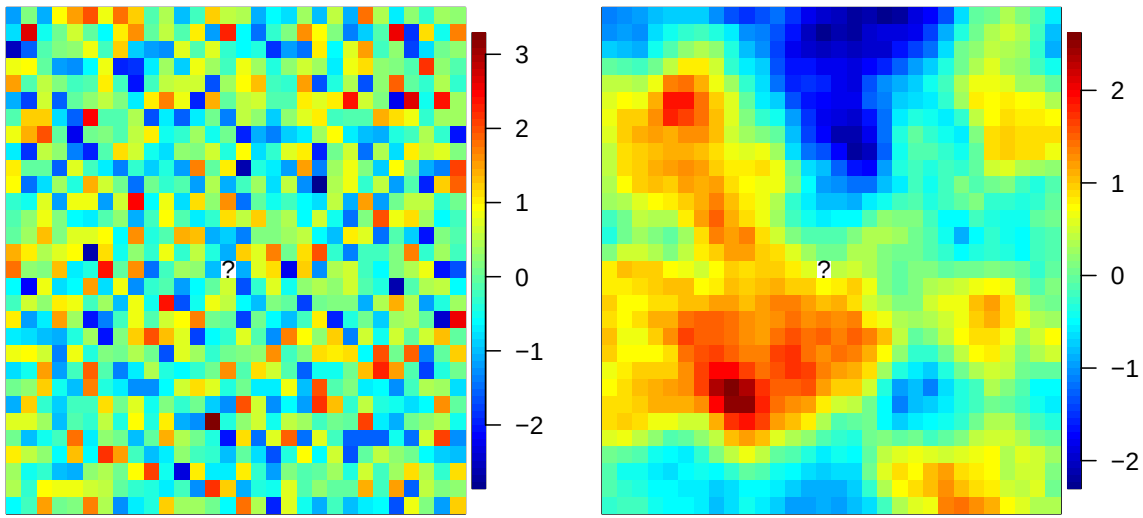
where

$$\begin{aligned} m_{Y_0|\boldsymbol{Y}=\boldsymbol{y}} &= m_0 + k^{\mathrm{T}}\Sigma^{-1}\left(\boldsymbol{y}-\boldsymbol{m}\right) \\ \sigma^2_{Y_0|\boldsymbol{Y}=\boldsymbol{y}} &= \sigma_0^2 - k^{\mathrm{T}}\Sigma^{-1}k \end{aligned}$$

Next, we are going to revisit our toy examples

Toy Examples Revisited

For simplicity, we assume $m(\mathbf{s}) = 0$ for $\mathbf{s} \in \mathcal{S}$, the spatial covariance only depends on distance



$$m_{Y_0|\mathbf{Y}=\mathbf{y}} = 0 + k^{\mathrm{T}}\Sigma^{-1}(\mathbf{y} - \mathbf{0}), \quad \sigma^2_{Y_0|\mathbf{Y}=\mathbf{y}} = \sigma_0^2 - k^{\mathrm{T}}\Sigma^{-1}k$$

Spatially uncorrelated field:

- $m_{Y_0|\mathbf{Y}} = 0$
- $\sigma^2_{Y_0|\mathbf{Y}=\mathbf{y}} = \sigma_0^2$

Spatially correlated field:

- $m_{Y_0|\mathbf{Y}} = k^{\mathrm{T}}\Sigma^{-1}\mathbf{y}$
- $\sigma^2_{Y_0|\mathbf{Y}=\mathbf{y}} = \sigma_0^2 - k^{\mathrm{T}}\Sigma^{-1}k$

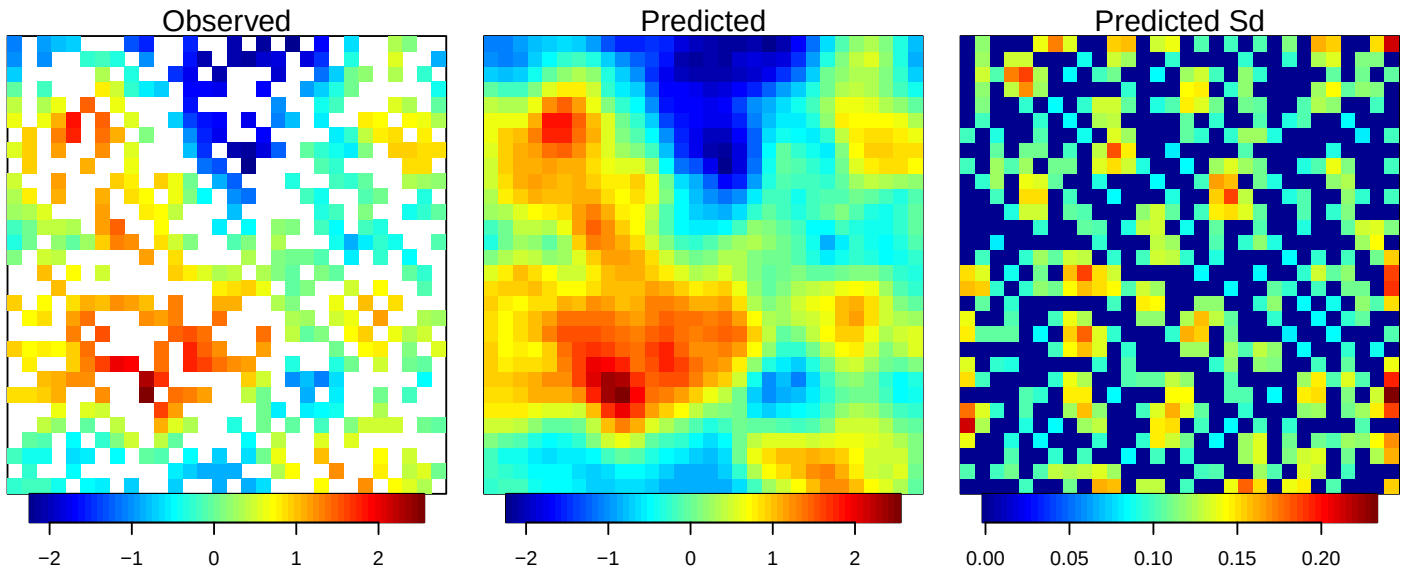
Interpolating Multiple Points in Space

In practice, we would like to predict the values at many locations. The Gaussian conditional distribution formula can still be used:

$$[\mathbf{Y}_0|\mathbf{Y}=\mathbf{y}] \sim \mathrm{N}\left(\boldsymbol{m}_{\mathbf{Y}_0|\mathbf{Y}=\mathbf{y}}, \Sigma_{\mathbf{Y}_0|\mathbf{Y}=\mathbf{y}}\right)$$

where

$$\boldsymbol{m}_{\mathbf{Y}_0|\mathbf{Y}=\mathbf{y}} = \boldsymbol{m}_0 + \boldsymbol{k}^{\mathrm{T}}\Sigma^{-1}(\mathbf{y}-\boldsymbol{m})$$
$$\Sigma_{\mathbf{Y}_0|\mathbf{Y}=\mathbf{y}} = \Sigma_0 - \boldsymbol{k}^{\mathrm{T}}\Sigma^{-1}\boldsymbol{k}$$



GP-Based Spatial Interpolation: Kriging

If $\{Y(\mathbf{s})\}_{\mathbf{s}\in\mathcal{S}}$ follows a GP, then

$$\begin{pmatrix}\mathbf{Y}_0\\ \mathbf{Y}\end{pmatrix} \sim \mathrm{N}\left(\begin{pmatrix}\boldsymbol{m}_0\\ \boldsymbol{m}\end{pmatrix}, \begin{pmatrix}\Sigma_0 & \boldsymbol{k}^{\mathrm{T}}\\ \boldsymbol{k} & \Sigma\end{pmatrix}\right)$$

We have

$$[\mathbf{Y}_0|\mathbf{Y}=\mathbf{y}] \sim \mathrm{N}\left(\boldsymbol{m}_{\mathbf{Y}_0|\mathbf{Y}=\mathbf{y}}, \Sigma_{\mathbf{Y}_0|\mathbf{Y}=\mathbf{y}}\right)$$

where

$$\begin{aligned}\boldsymbol{m}_{\mathbf{Y}_0|\mathbf{Y}=\mathbf{y}} &= \boldsymbol{m}_0 + \boldsymbol{k}^{\mathrm{T}}\Sigma^{-1}(\mathbf{y}-\boldsymbol{m})\\ \Sigma_{\mathbf{Y}_0|\mathbf{Y}=\mathbf{y}} &= \Sigma_0 - \boldsymbol{k}^{\mathrm{T}}\Sigma^{-1}\boldsymbol{k}\end{aligned}$$

Question: what if we don't know $m(s;\boldsymbol{\beta}), c(h;\boldsymbol{\theta})$?

$\Rightarrow$ We need to estimate the mean and covariance from the data $\boldsymbol{y}$.

Outline

Kalman Recursions for Filtering, Forecasting, and Smoothing

Estimating the State-Space Model Parameters

Gaussian Process Spatial Model

Spatial Interpolation

Parameter estimation

A Case Study of Paraná State Precipitation Data

Variogram, Semivariogram, and Covariance

For two spatial locations s_i and s_j, the semivariogram is defined as

$$\gamma(s_i, s_j) = \frac{1}{2}\,\mathrm{Var}\{Y(s_i) - Y(s_j)\}.$$

Under second-order stationarity,

$$E\{Y(s)\} = \mu, \qquad \mathrm{Cov}\{Y(s_i), Y(s_j)\} = C(s_i - s_j).$$

Therefore,

$$2\gamma(s_i, s_j) = E\left[\{Y(s_i) - Y(s_j)\}^2\right]$$

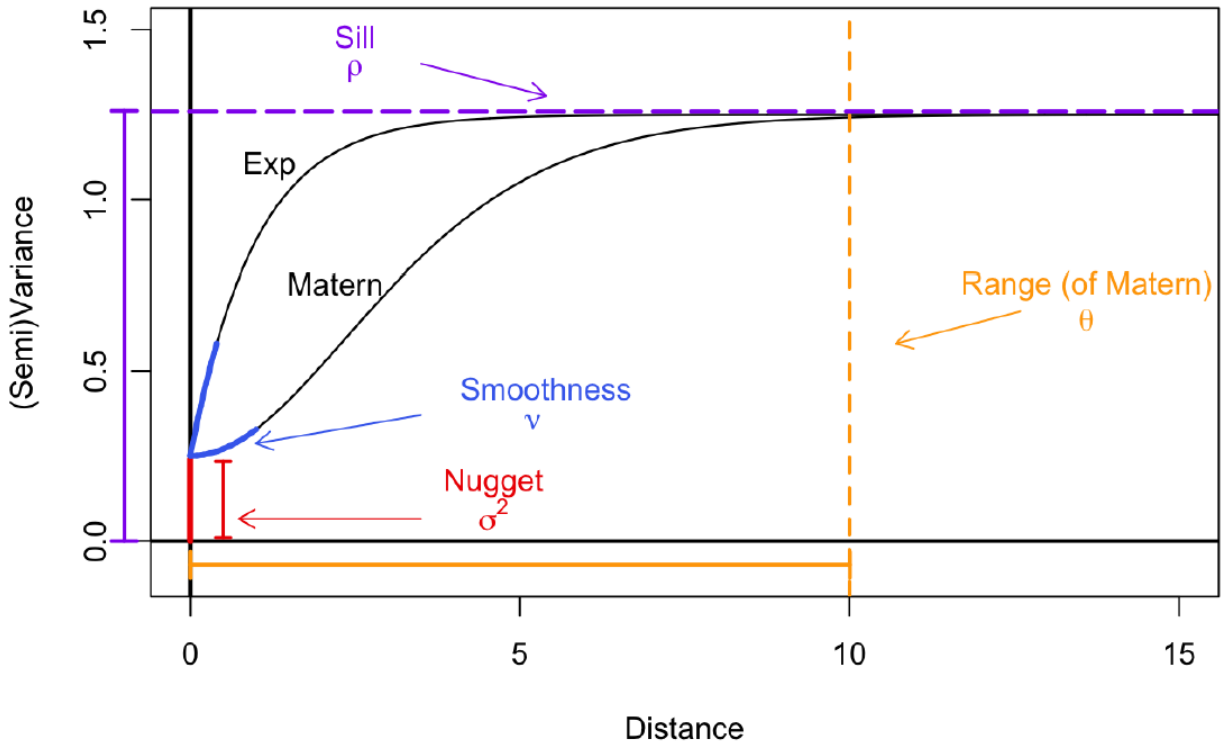
If the process is also isotropic, then the semivariogram depends only on distance $h = \|s_i - s_j\|$, and

$$\gamma(h) = C(0) - C(h)$$

Interpretation

Small semivariogram values indicate strong similarity between nearby locations

Semivariogram $\left\{\frac{1}{2}\mathrm{Var}\left(\varepsilon\left(\mathbf{s}_i\right)-\varepsilon\left(\mathbf{s}_j\right)\right)\right\}_{i,j}$



Source: `fields` vignette by Wiens and Krock, 2019

- **Nugget**: error / microscale variation
- **Sill = Nugget + Partial Sill**: total variance
- **Range**: dependence scale
- **Smoothness**: process roughness

Key idea: The empirical semivariogram guides covariance model selection

Estimation: Weighted Least Squares

A common approach is to fit a parametric semivariogram model

$$\gamma(h; \boldsymbol{\theta})$$

to the empirical semivariogram

$$\hat{\gamma}(h_u)$$

The weighted least squares estimator solves

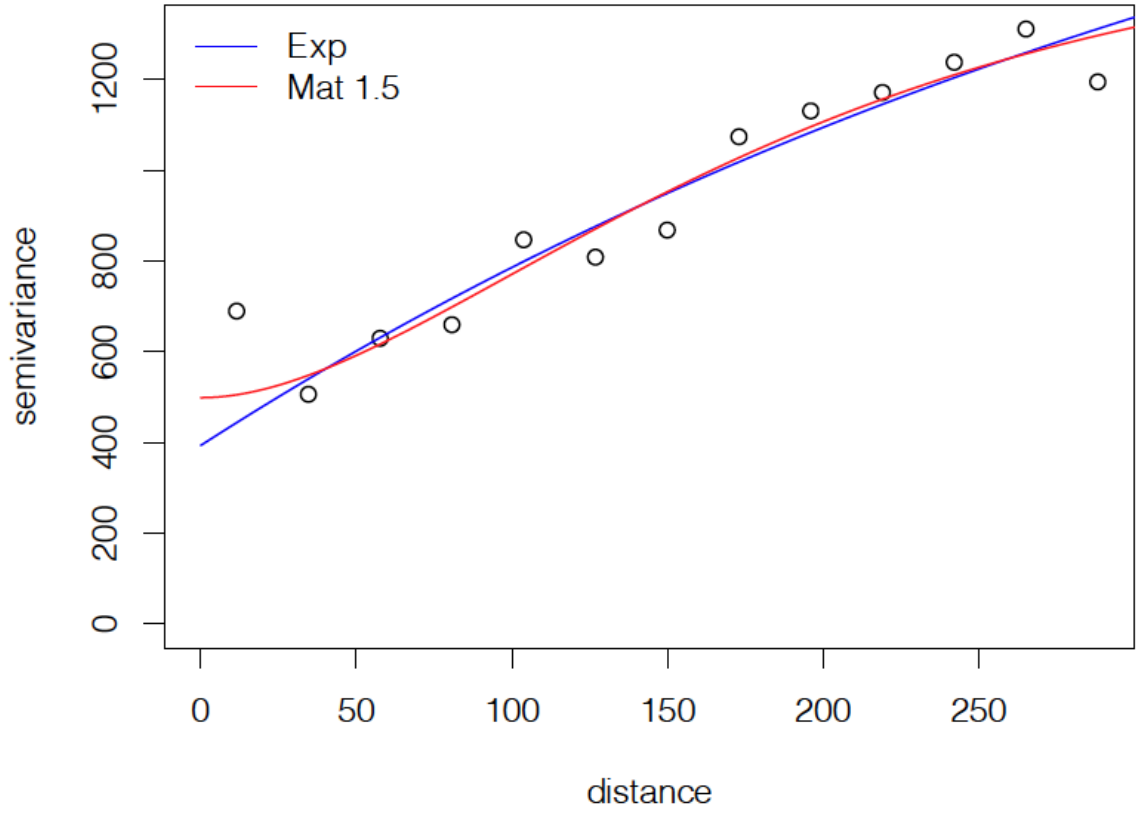
$$\arg\min_{\boldsymbol{\theta}} \sum_{u \in U} \frac{N(h_u)}{\{\gamma(h_u; \boldsymbol{\theta})\}^2} \left[\hat{\gamma}(h_u) - \gamma(h_u; \boldsymbol{\theta})\right]^2$$

Why These Weights?

The empirical semivariogram is more precise when:

- more location pairs are available ($N(h_u)$ is large);
- the semivariogram value is small, since the variability of $\hat{\gamma}(h_u)$ typically increases with $\gamma(h_u)^2$.

Fitting Semivariogram Models



Takeaway

Fitted semivariogram models provide covariance parameters for kriging. We next consider MLE, which accounts for the full spatial dependence structure.

Maximum Likelihood Estimation

Suppose

$$Y(s) = x(s)^T \beta + \epsilon(s),$$

where $\varepsilon(s)$ is a Gaussian process with covariance

$$\mathrm{Cov}\{\varepsilon(s_i), \epsilon(s_j)\} = C_\theta(s_i, s_j).$$

For observed locations $s_1, \ldots, s_n$,

$$\mathbf{Y} = \big(Y(s_1), \ldots, Y(s_n)\big)^T$$

has a multivariate normal distribution

$$\mathbf{Y} \sim N(X\beta, \Sigma_\theta),$$

where

$$\Sigma_\theta(i, j) = C_\theta(s_i, s_j).$$

MLE estimates β and θ by maximizing the multivariate normal likelihood

Profile Likelihood

For a fixed covariance parameter vector $\boldsymbol{\theta}$, the generalized least squares estimator is

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\theta}) = (X^T \Sigma_{\boldsymbol{\theta}}^{-1} X)^{-1} X^T \Sigma_{\boldsymbol{\theta}}^{-1} \mathbf{y}.$$

Substituting $\hat{\boldsymbol{\beta}}(\boldsymbol{\theta})$ into the likelihood gives the profile log-likelihood:

$$\ell(\boldsymbol{\theta}; \mathbf{y}) \propto -\frac{1}{2} \log |\Sigma_{\boldsymbol{\theta}}| - \frac{1}{2} \mathbf{y}^T P(\boldsymbol{\theta}) \mathbf{y},$$

where

$$P(\boldsymbol{\theta}) = \Sigma_{\boldsymbol{\theta}}^{-1} - \Sigma_{\boldsymbol{\theta}}^{-1} X (X^T \Sigma_{\boldsymbol{\theta}}^{-1} X)^{-1} X^T \Sigma_{\boldsymbol{\theta}}^{-1}.$$

Goal

Maximize the profile likelihood to estimate the covariance parameters.

Remarks on Likelihood-based estimation

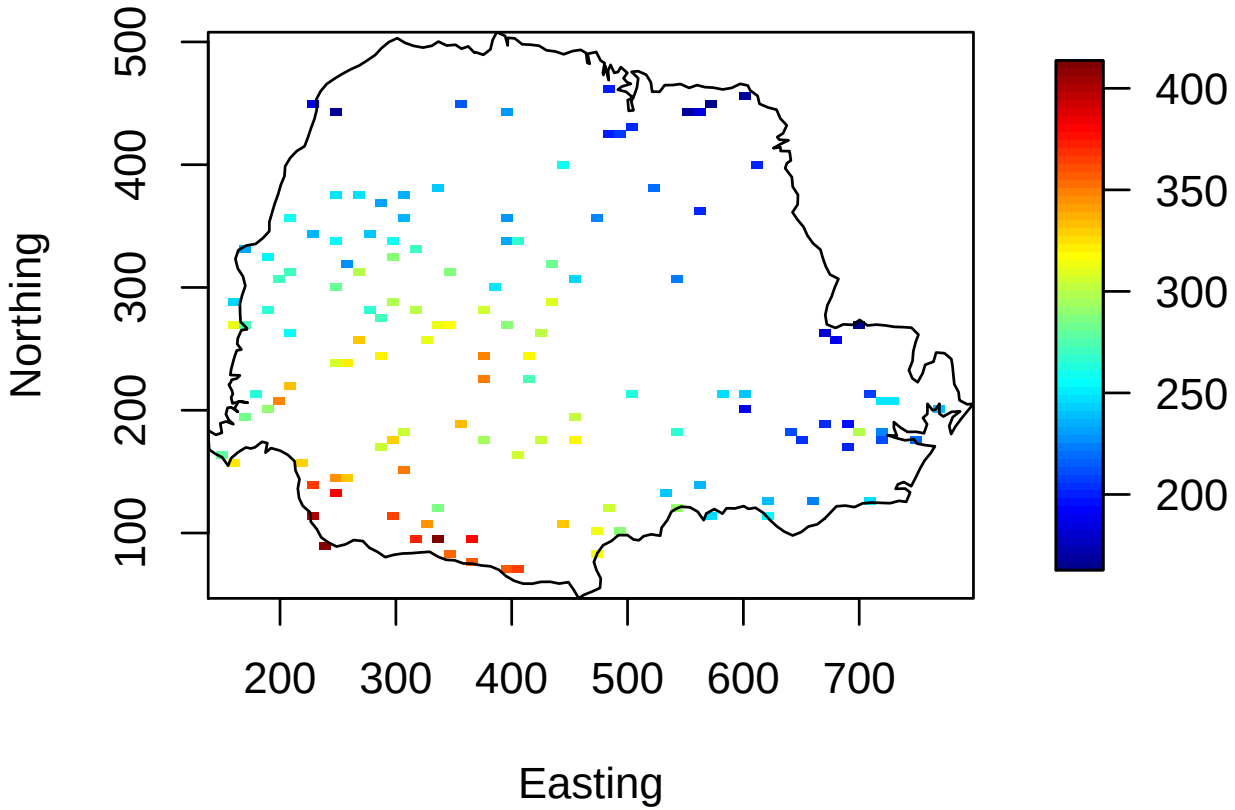
- Maximizing $\ell_n(\boldsymbol{\theta}; \mathbf{y})$ involves solving a constrained nonlinear optimization problem, necessitating numerical methods for obtaining ML estimates.

- Alternatively, Restricted (or residual) maximum likelihood (REML) can be employed.

- Likelihood-based estimation poses computational challenges with large spatial datasets, primarily due to the significant computational complexity, requiring $\mathcal{O}(n^3)$ operations and $\mathcal{O}(n^2)$ memory.

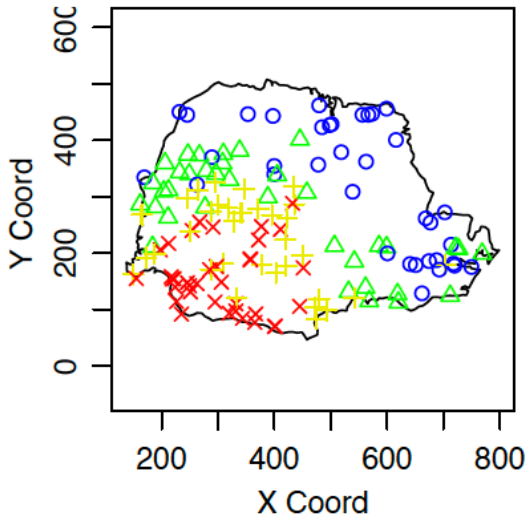
Paraná State Precipitation Data

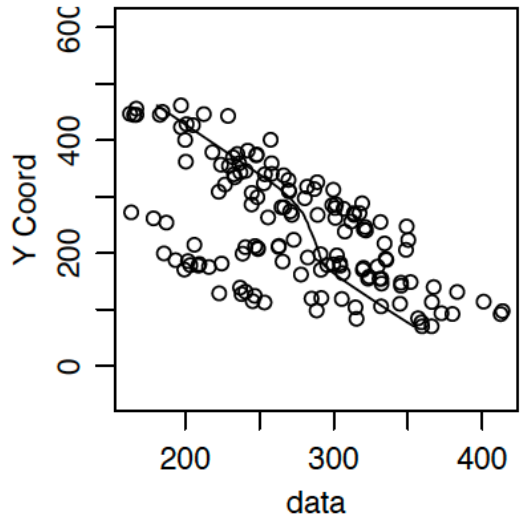
We look at the average winter (May-June, dry season) rainfall at 143 locations throughout Paraná, Brazil

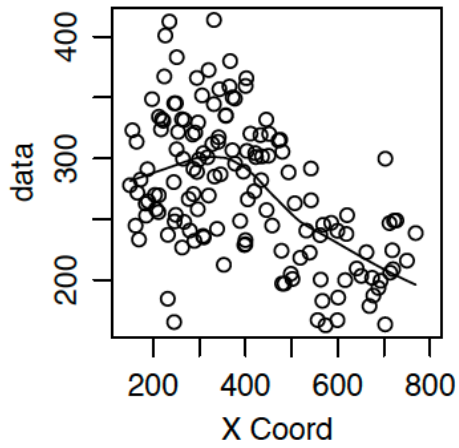


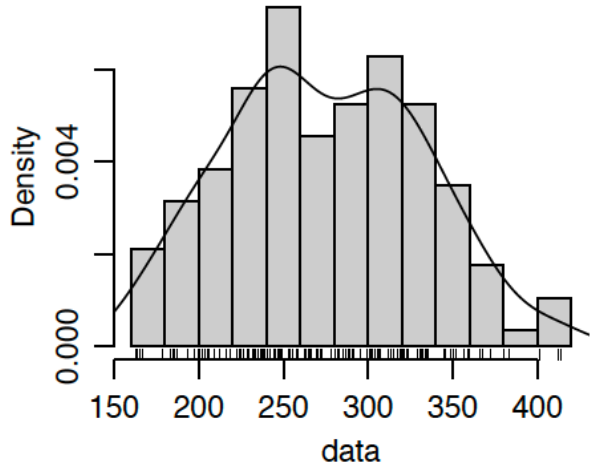
Goal: To interpolate the values in the spatial domain

Exploratory Data Analysis



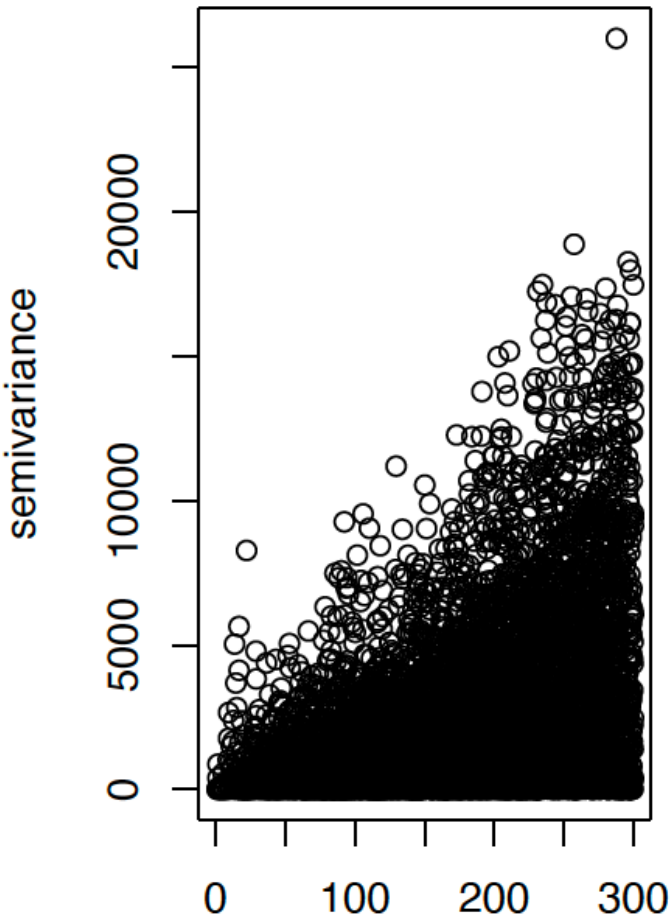


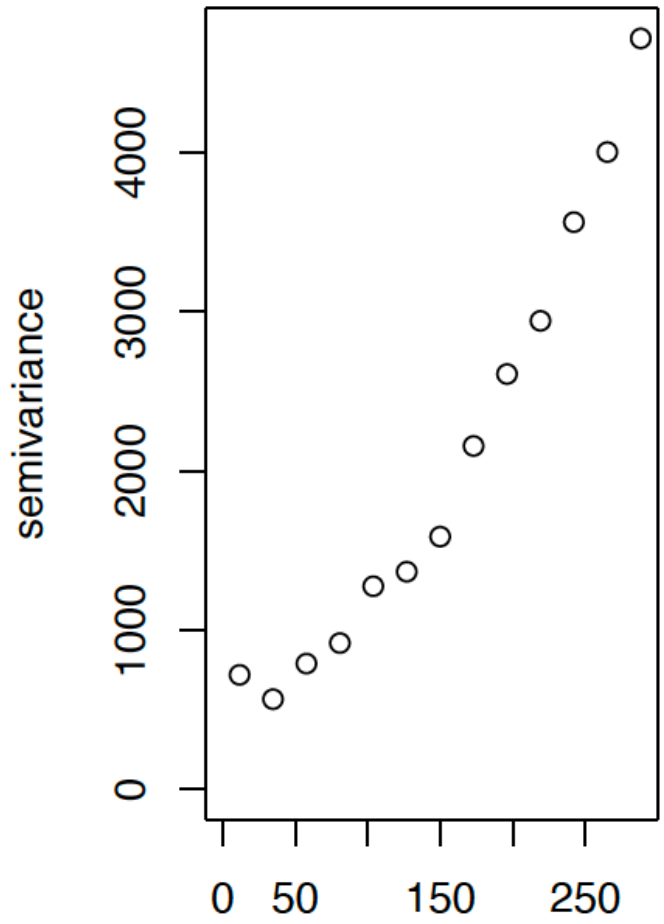




A linear trend in space (both longitude and latitude) may be suitable to characterize the large-scale spatial trend

Variogram Analysis



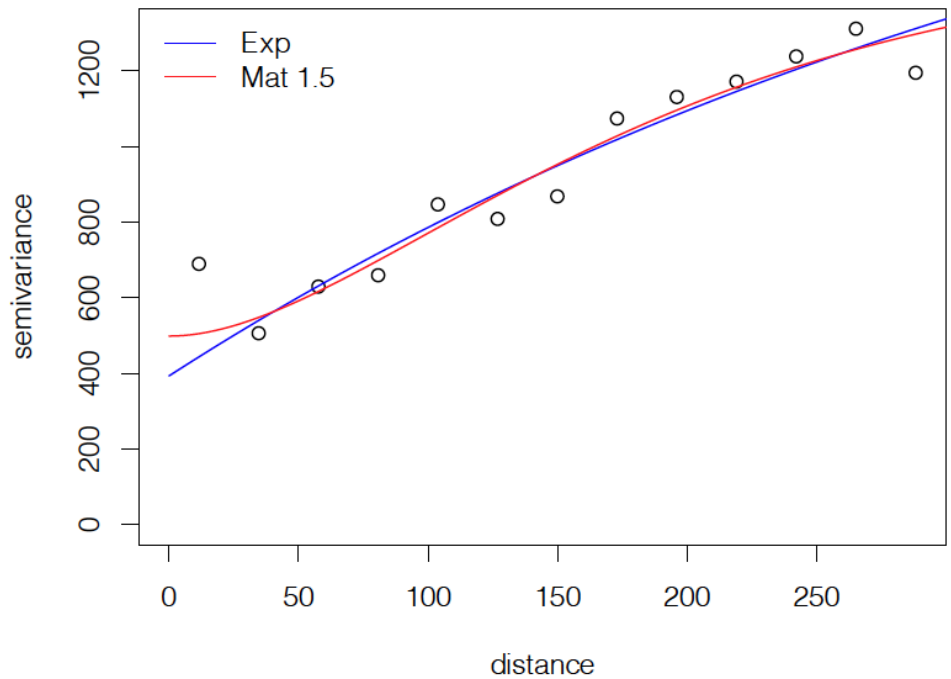


Interpretation

Nearby locations have more similar precipitation values than distant locations, suggesting positive spatial dependence

Estimating Spatial Covariance via Variogram

```
parana.vtfit.exp <- variofit(parana.variot)
parana.vtfit.mat1.5  <- variofit(parana.variot, kappa = 1.5)
```



We fit candidate semivariogram models to the empirical variogram

The fitted variogram provides initial estimates of covariance parameters for maximum likelihood estimation

Maximum Likelihood Estimation of Paraná Rainfall

```
(parana.ml1 <- likfit(parana, trend = "1st", ini = c(1000, 50), nug = 100))
```

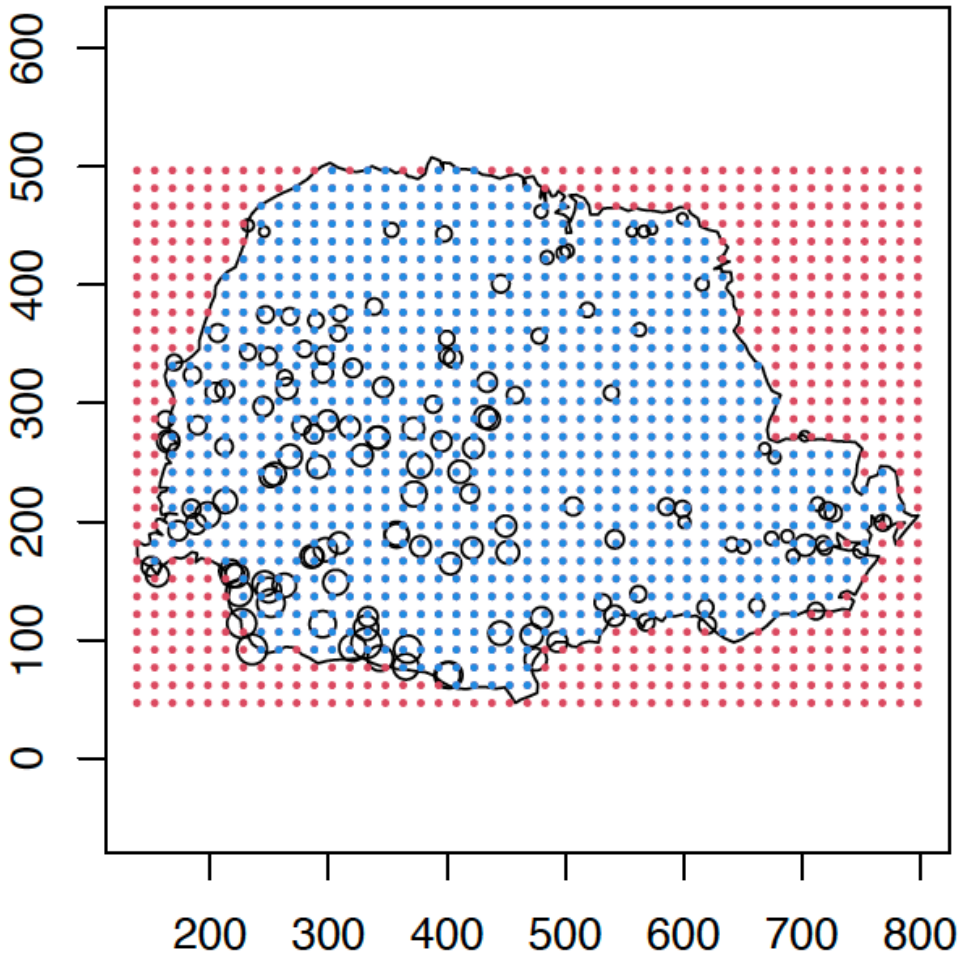
```
## ---------------------------------------------------------------
## likfit: likelihood maximisation using the function optim.
## likfit: Use control() to pass additional
##          arguments for the maximisation function.
##         For further details see documentation for optim.
## likfit: It is highly advisable to run this function several
##         times with different initial values for the parameters.
## likfit: WARNING: This step can be time demanding!
## ---------------------------------------------------------------
## likfit: end of numerical maximisation.

## likfit: estimated model parameters:
##      beta0      beta1      beta2      tausq    sigmasq        phi
## "416.4984" " -0.1375" " -0.3997" "385.5180" "785.6904" "184.3863"
## Practical Range with cor=0.05 for asymptotic range: 552.3719
##
## likfit: maximised log-likelihood = -663.9
```

Next Step

Use the estimated mean and covariance parameters to perform spatial interpolation.

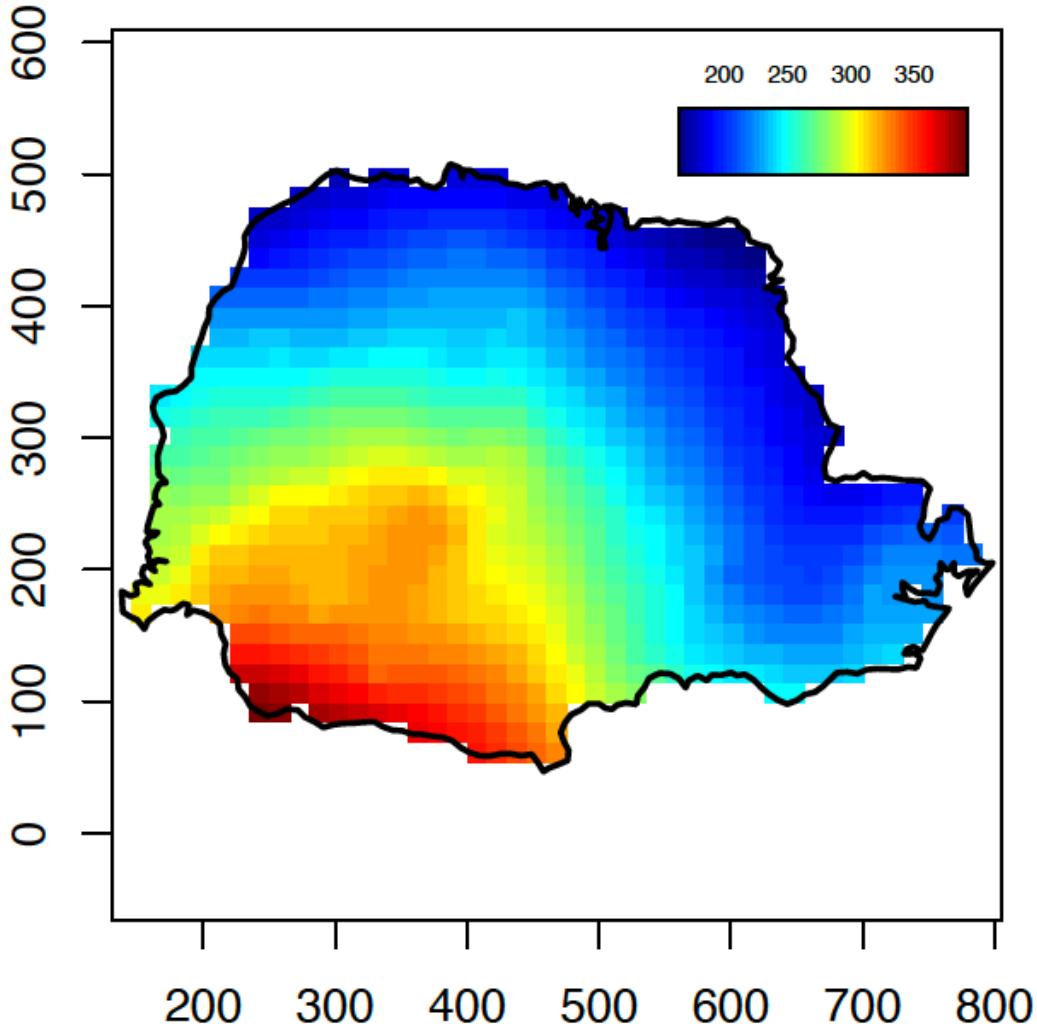
Setting Up the Spatial Grids for Prediction



Practical Step

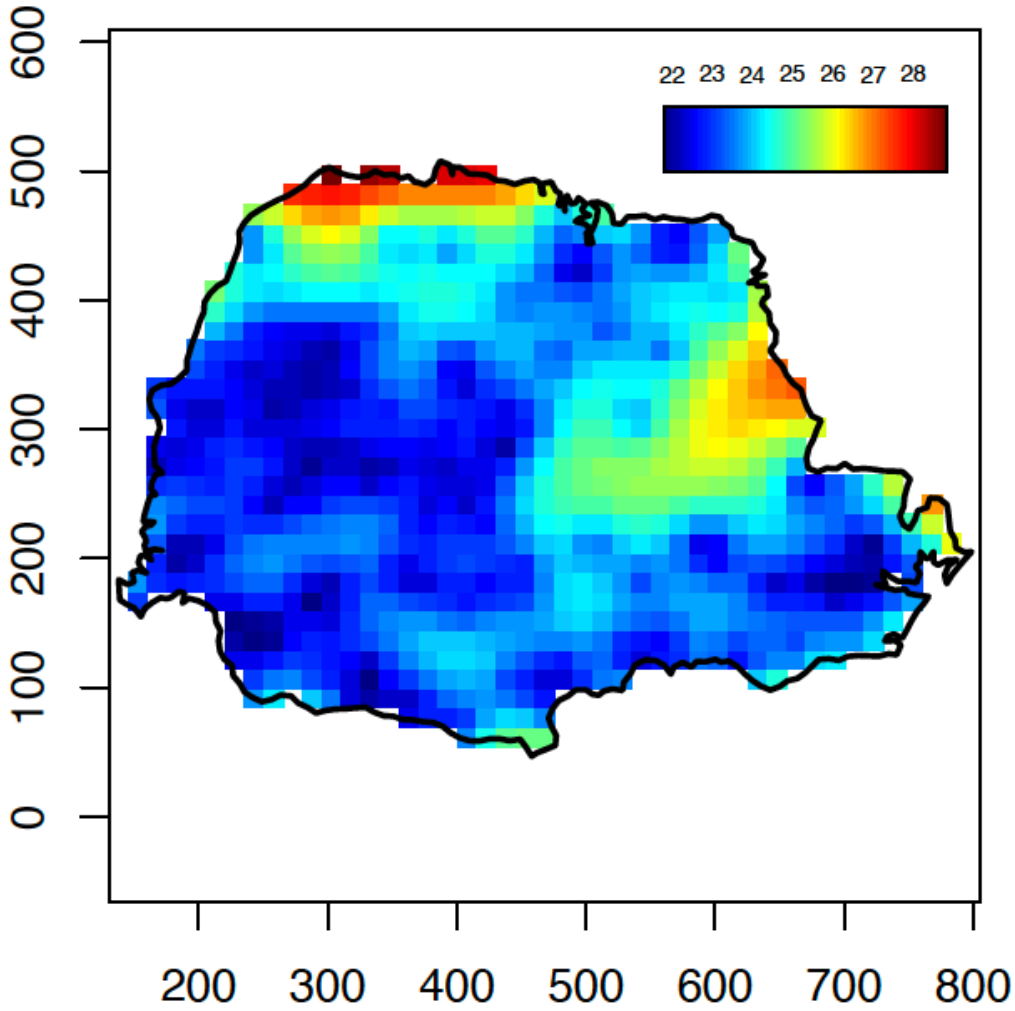
Kriging is applied at each grid point to produce a spatial prediction surface

Spatial Predicted Map



The smooth spatial surface reflects both the fitted mean trend and the estimated spatial covariance structure

Prediction Uncertainty Map



Prediction uncertainty is generally smaller near observed monitoring stations and larger in regions with sparse observations or near the boundary